

Caylee Spivey

✉ caylee.spivey@uconn.edu 🔗 caylee-spivey.github.io

Education

- University of Connecticut**, PhD in Mathematics *Expected: Spring 2027*
- **Advisor:** Keith Conrad
 - **Thesis:** “Partial Sums of p -adic Logarithms at Torsion Points”
- University of North Georgia**, BS in Mathematics *Fall 2016 - Spring 2020*
- **Senior Project:** “When Additive and Multiplicative Inverses are the Same”

Research Interests

Algebraic number theory, p -adic analysis, scholarship of teaching and learning

Awards

- Teaching Innovation Fellowship** *Fall 2026 - Spring 2027*
Mathematics Department, University of Connecticut
- Summer Disseration Fellowship** *Summer 2026*
Graduate School, University of Connecticut
- Connie Strange Graduate Community Award** *April 2026*
Mathematics Department, University of Connecticut
- Math and Cobb Travel Fund** *January 2026*
Álvaro Lozano-Robledo
- National Science Foundation Fellow** *Fall 2025 - Summer 2026*
Transdisciplinary Convergence in Educational Neuroscience
Doctoral Training Program, University of Connecticut
- Louis J. DeLuca Award for Excellence in Teaching** *April 2025*
Mathematics Department, University of Connecticut
- Walter J. Carpenter Mathematics Award** *April 2019*
Mathematics Department, University of North Georgia

Publications

- The Distribution of k -free Ideals in Ray Class Groups** *August 2026*
Joint with A. Barquero-Sanchez, J. Heimrath, B. Sing, N. Sirotli, M. Wijaya
Submitted. [arXiv](#) [🔗](#)
- Environmental Underreporting in Deaf Education: A Scoping Review and Proposal** *June 2026*
Joint with A. Bobrycki, M. Chappell.
Accepted. To appear in the Journal of Deaf Studies and Deaf Education.
- When Additive and Multiplicative Inverses are the Same** *May 2023*
Joint with K. Briggs.
[Mathematics Magazine](#) [🔗](#), 96(3), 299–307.

Teaching

COURSES TAUGHT

- Primary Instructor, Teaching Innovation Fellow, UConn Mathematics** *Fall 2026*
- MATH 1132Q: Calculus II (one class)

Math Coordinator, Center for Access & Postsecondary Success, UConn	<i>Summer 2026</i>
◦ Held weekly meetings with mathematics instructors and TAs, communicated regularly with CAPS administrators	
Primary Instructor, Center for Access and Postsecondary Success, UConn	<i>Summer 2025, 2026</i>
◦ MATH 1030Q: Elementary Discrete Mathematics (one class)	
Primary Instructor, UConn Mathematics	
◦ MATH 1071Q: Calculus for Business and Economics (two classes)	<i>Fall 2024, Spring 2025</i>
◦ MATH 1020Q: Problem Solving (two classes)	<i>Fall 2023, Spring 2024</i>
Teaching Assistant, UConn Mathematics	
◦ MATH 1071Q: Calculus for Business and Economics (online office hours)	<i>Summer 2026</i>
◦ MATH 1132Q: Calculus II (online office hours)	<i>Summer 2023, 2024</i>
◦ MATH 1131Q: Calculus I (two discussion sections)	<i>Fall 2022, Spring 2023</i>

COURSE DEVELOPMENT

MATH 1070Q: Mathematics for Business and Economics	<i>May 2026</i>
◦ Created online question banks in Blackboard to be used in homework assignments and exams.	
MATH 1071Q: Calculus for Business and Economics	<i>January 2025</i>
◦ Collaboratively developed guided notes for use by all graduate student instructors.	
MATH 1020Q: Problem Solving	<i>Summer 2023</i>
◦ Collaborated with other instructors to develop projects to replace traditional assessments.	

Conference Talks

Connecticut Summer School in Number Theory	<i>June 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“Counting k -free Ideals in Generalized Arithmetic Progressions”	
Rutgers Mathematics End of Year Conference	<i>May 2026</i>
<i>Rutgers University</i>	<i>New Brunswick, NJ</i>
“Mathematics and Educational Neuroscience: Bridging the Gap Between Research and Practice”	
Graduate Research Forum	<i>May 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“The 2-power divisibility of $2 + 2^2/2 + \cdots + 2^n/n$ ”	
Mathematics Continued Conference (Invited)	<i>March 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“Removing Limits from Calculus: A Discrete Approach”	
Joint Mathematics Meetings (Invited)	<i>January 2026</i>
<i>AMS Special Session on Rethinking Number Theory: Highlighting the Research and Discussions of the RNT Workshops</i>	<i>Washington, DC</i>
“Counting k -free Ideals in Arithmetic Progressions in Number Fields”	
Women in Mathematics in New England	<i>September 2025</i>
<i>Smith College</i>	<i>Northampton, MA</i>
“Discrete Calculus and its Applications”	
MAA MathFest	<i>August 2021 (Online)</i>
“Computational Explorations in Abstract Algebra” (presented with K. Briggs)	
MAA Southeastern Sectional Meeting	<i>March 2021 (Online)</i>
“When Additive & Multiplicative Inverses are the Same” (presented with K. Briggs)	

Seminar Talks

UConn Math Club <i>University of Connecticut</i> “The Mathematics of Democracy”	<i>October 2026</i> <i>Storrs, CT</i>
UConn Algebra Seminar <i>University of Connecticut</i> “The Distribution of k -free Ideals in Generalized Arithmetic Progressions”	<i>April 2026</i> <i>Storrs, CT</i>
UConn TalkShop (BrownBag Seminar) <i>University of Connecticut</i> “What Do Mathematicians Actually Do?”	<i>April 2026</i> <i>Storrs, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “Topological Groups and Their Applications”	<i>March 2026</i> <i>Storrs, CT</i>
UConn Teaching Workshop <i>University of Connecticut</i> “How to TRANSCEND Math Research: An Introduction to Educational Neuroscience”	<i>November 2025</i> <i>Storrs, CT</i>
University of Hartford Undergraduate Mathematics Colloquium <i>University of Hartford</i> “Fermat’s Last Theorem for Polynomials”	<i>October 2025</i> <i>West Hartford, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “When Polynomials Make Life Easier: Fermat’s Last Theorem, Catalan’s Conjecture, and More”	<i>October 2025</i> <i>Storrs, CT</i>
UConn Math Club <i>University of Connecticut</i> “Fermat’s Last Theorem for Polynomials”	<i>October 2025</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “Number Fields and the LMFDB”	<i>September 2025</i> <i>Storrs, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “Discrete Calculus and its Applications”	<i>February 2025</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “The Topology of Dual Groups”	<i>February 2025</i> <i>Storrs, CT</i>
UConn Math Club <i>University of Connecticut</i> “Calculus of Finite Differences”	<i>November 2024</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “Topological Groups”	<i>February 2025</i> <i>Storrs, CT</i>

Conferences/Workshops Attended

MAA MathFest <i>Boston Marriott Copley Place</i>	<i>August 2026</i> <i>Boston, MA</i>
Connecticut Summer School in Number Theory <i>University of Connecticut</i>	<i>June 2022, 2024, 2026</i> <i>Storrs, CT</i>

Rutgers Mathematics End of Year Conference <i>Rutgers University</i>	<i>May 2026</i> <i>New Brunswick, NJ</i>
Joint Mathematics Meetings <i>Walter E. Washington Convention Center</i>	<i>January 2026</i> <i>Washington, DC</i>
Maine-Québec Number Theory Conference <i>University of Maine</i>	<i>October 2025</i> <i>Orono, ME</i>
Rethinking Number Theory <i>AIM Research Community</i>	<i>June 2025</i> <i>Online</i>
Project: Arithmetic Functions over Number Fields	
Séminaire de Mathématiques Supérieures 2025: An Introduction to Recent Trends in Commutative Algebra <i>The Fields Institute</i>	<i>June 2025</i> <i>Toronto, ON</i>
Workshop: Murmurations in Arithmetic Geometry and Related Topics <i>Simons Center for Geometry and Physics</i>	<i>November 2024</i> <i>Stony Brook, NY</i>

Poster Presentations

MAA MathFest <i>Boston Marriott Copley Place</i>	<i>August 2026</i> <i>Boston, MA</i>
“How to TRANSCEND Math Research: An Introduction to Educational Neuroscience”	
NSF Research Traineeship Annual Meeting <i>Colorado School of Mines</i>	<i>March 2026</i> <i>Golden, CO</i>
“TRANSCEND: TRANSDisciplinary Convergence in Educational Neuroscience Doctoral Training Program”	
NSF Research Traineeship Annual Meeting <i>Colorado School of Mines</i>	<i>March 2026</i> <i>Golden, CO</i>
“School & Classroom Factors Influencing Literacy & Mathematics Achievement In Deaf & Hard Of Hearing Students”	

Mentoring

Graduate Student Mentor Network, Coordinator <i>University of Connecticut</i>	<i>Fall 2024 - present</i> <i>Storrs, CT</i>
Paired first-year mathematics graduate students with returning students. Hosted informational panel during orientation week. Organized social events for mentors and mentees.	
Directed Reading Program, Mentor <i>University of Connecticut</i>	<i>Fall 2025 - Spring 2026</i> <i>Storrs, CT</i>
Mentor to Erica Davies (2 semesters). Topics: Elliptic Curves, SageMath, Applications to Cryptography	

Professional Service/Activities

AWM Graduate Student Chapter, Treasurer <i>University of Connecticut</i>	<i>Fall 2023 - present</i> <i>Storrs, CT</i>
Organized social and professional development activities for graduate students in the mathematics department.	
UConn Mathematics Teaching Workshop, Member <i>University of Connecticut</i>	<i>Fall 2022 - present</i> <i>Storrs, CT</i>
Meet weekly with graduate students and faculty to discuss pedagogy, teaching experiences, and research topics of interest to the group.	
Number Theory Reading Group, Member <i>University of Connecticut</i>	<i>Fall 2022 - present</i> <i>Storrs, CT</i>
Topics covered: modular forms, ranks of elliptic curves, Tate’s thesis, the LMFDB, L-functions.	

MATHCOUNTS Eastern Connecticut Chapter Competition, Volunteer*University of Connecticut**February, 2003 - 2026**Storrs, CT*

- Served as a grader, registration coordinator, and scoring supervisor. MATHCOUNTS is a mathematics competition for middle school students.

Mathematics Continued Conference, Speaker/Panelist/Volunteer*University of Connecticut**Oct. '22, Mar. '23 - '26**Storrs, CT*

- Served as concurrent session moderator and graduate school panelist. Gave a talk in 2026. The Mathematics Continued Conference is aimed at undergraduate students who are interested in continuing to study mathematics after graduating.

Prelim Preparation Workshop, Organizer*University of Connecticut**November 2025**Storrs, CT*

- Invited professors and senior graduate students to give advice to new students preparing for preliminary exams.

Women in Mathematics in New England, Panelist*Smith College**September 2022, 2025**Northampton, MA*

- Served on the graduate school informational panel.

Becoming an Instructor of Record Panel, Organizer*University of Connecticut**October 2024**Storrs, CT*

- Organized a panel of senior graduate students to discuss what it is like becoming a primary instructor.

AMS Graduate Student Chapter, Secretary*University of Connecticut**Fall 2022 - Spring 2024**Storrs, CT*

- Organized social and professional development activities for graduate students in the mathematics department.

UConn Math Club, Panelist*University of Connecticut**April 2023**Storrs, CT*

- Served on graduate school informational panel.

Math Club President*University of North Georgia**Fall 2019 - Spring 2021**Dahlonega, GA*

- Organized social events, mathematics enrichment activities, and a seminar for faculty to present their research interests to students.

Professional Memberships

- American Mathematical Society
- Association for Women in Mathematics
- Mathematical Association of America

Programming Languages

- Magma
- MATLAB
- Python
- SageMath